

The Ecological Effect of Drought on the Macroinvertebrate Fauna of a Small Intermittent Stream – An Example from the Czech Republic

key words: macrozoobenthos, abundance, taxa richness, hydrology

Abstract

The impact of an unstable hydrological regime on macroinvertebrates in an intermittent stream (with periodic summer droughts in particular stretches) was investigated in two successive years with differing climatic conditions. Ten series of macroinvertebrate samples were taken from two sampling sites between April 2002 and October 2003. Total abundance, taxa richness, main faunal groups, and life strategies were assessed in order to describe the changes in the macroinvertebrate assemblages. Periodic dry periods caused a decrease in macroinvertebrate density in the intermittent stretch, whereas they had no negative influence on number of taxa. Before the drought, the density of permanent fauna decreased and the number of r-strategists rose and stayed at a high level in first samples in the after-drought period.

1. Introduction

Drying up of streams is not a common phenomenon in natural conditions in Central Europe. Natural low flows are more frequent in Australia (BROCK *et al.*, 2003; McMAHON and FINLAYSON, 2003), South and North America (MILLER and GOLLADAY, 1996; DEL ROSARIO and RESH, 2000; SMITH *et al.*, 2001; COVICH *et al.*, 2003) and the south of Europe (PIRES *et al.*, 2000), generally in karst and chalk streams (WRIGHT *et al.*, 2004). Low flows can also be caused by anthropogenic activities (river regulation, dams, *etc.*) and these changes in ecosystems are more rapid and dramatic (WILLIAMS and FELTMATE, 1994; KUBÍČEK *et al.*, 1999; McMAHON and FINLAYSON, 2003).

In this study, two types of habitats within stream were distinguished according to the natural periodicity of their discharge conditions. The middle stretch of the Gránický brook has intermittent flow but the lower stretch is perennial. The intermittent stream type dries up periodically and is characterised by greater frequency and variability of disturbances. The development of macroinvertebrate assemblages in such a “stream type” is seasonal and predictable. Perennial streams are more common in Central Europe and drying up of this stream type is unpredictable, and only occurs in extremely dry years. As a result, such drought events are less frequent (MILLER and GOLLADAY, 1996; DEL ROSARIO and RESH, 2000; HANSEN, 2001).

Drought conditions lead to changes in abiotic parameters such as decreases in dissolved oxygen and phosphorus concentrations and increases in conductivity, water temperature, and nitrogen concentrations (WILLIAMS and FELTMATE, 1994; HUMPHRIES and BALDWIN, 2003).

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The drying up of the streambed impacts on all stream organisms. Particularly important are the duration and intensity of the dry periods (depending on the depth of the water table under the streambed surface, the length of the dry stretch, *etc.*).

During the drying up process habitat diversity decreases. This phenomenon is closely associated with the lack of food sources and induces stronger intra- and interspecific competition (WILLIAMS and FELTMATE, 1994; HUMPHRIES and BALDWIN, 2003). Organisms can respond to changes in different ways. Some can survive using resistant eggs, seeds or spores (BROCK *et al.*, 2003) whilst other, *e.g.*, insect species, can reduce their life cycle duration. Another possibility is moving to refugia (*e.g.*, downstream drift, active upstream movement or downward movement in the hyporheal). Drift is used, for example, by insect larvae and the crustacean *Gammarus fossarum* KOCH, 1836. Fish commonly use active upstream movement (MAGOULICK and KOBZA, 2003; MATTHEWS and MARSH-MATTHEWS, 2003) but upstream macroinvertebrate migration is also known (SÖDERSTRÖM, 1987). The hyporheal can also serve as a refugium for invertebrates with small, hard bodies (THORP and COVICH, 1991; WILLIAMS and FELTMATE, 1994; BRETSCHKO, 1995; RULÍK, 1995; DEL ROSARIO and RESH, 2000; FOWLER, 2004) but this possibility mainly depends on the porosity of the substratum and the permeability of the stream bed.

The aims of this study were to find out if there were (i) differences in taxon composition of macroinvertebrate community of intermittent and perennial stream type, (ii) changes in life strategies of macroinvertebrates induced by drought, and (iii) structural changes of the assemblage caused by drought.

2. Methods

2.1. Study Area

This study was carried out in the Gránický brook (Fig. 1), which is situated in the south of Moravia (Czech Republic). It is a tributary of the River Dyje (Thaya), in the Danube Catchment, and it is situated on the boundary of Hercynian and Pannonian subprovinces (CULEK, 1996).

This region is one of the driest parts of the Czech Republic (average annual rainfall is only 665 mm) and the middle part of the brook dries up every summer due to climatic, geological and geomorpho-

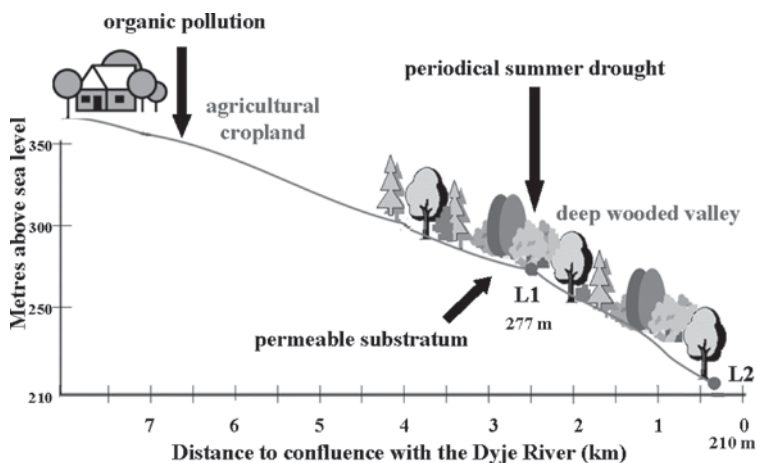


Figure 1. Longitudinal profile of the Gránický brook: L1 – intermittent site; L2 – perennial site.

logical conditions. Annual precipitation in 2002 was higher than the long-term average (691 mm), while in 2003 it was very low (402 mm).

The total length of the brook is 13 km and its catchment area is 20.5 km². In the section studied, the brook has a high slope and a cobble-gravel-sand substratum. The watercourse here meanders through a deep wooded valley with no organic loading or human impact. Its high conductivity is natural in origin and results from the geological conditions (sandy clay sediments and schistose biotic granites with high concentrations of sulphates).

Two sites of different character were observed in two successive years: (i) intermittent (middle stretch, altitude 277 m), and (ii) perennial (lower stretch, altitude 210 m). The intermittent site was 1.9 km upstream of the perennial site, which is situated 300 m above the confluence of the brook with the River Dyje (Thaya). The hydrological regime of the middle intermittent stretch is typically unstable, with periodic summer drying up. In contrast, the lower perennial stretch only dries up in extremely dry years, such as in 2003. The dry period lasted for approximately two weeks in 2002, when only the intermittent site dried up. In the following year, the whole brook dried up but the duration of the drought was different at each site. The intermittent site was dry for more than 3 months, from mid July to the end of the investigation in October. However, the perennial site dried for only six weeks, from mid August to the beginning of October, when flow resumed in all reaches except the intermittent site.

2.2. Sampling Methods

Ten series of samples were taken from April 2002 to October 2003. Samples were taken only when the site was flooded with water.

Benthic macroinvertebrates were sampled using the PERLA method (KOKEŠ *et al.*, 2006) that comprised semi-quantitative, multihabitat 3 minute kick samples gathered with a hand net (25 × 25 cm aperture, mesh size 0.5 mm, sack length 75 cm). Samples were collected from an approximately 20 m long, representative stretch of the site and each type of mesohabitat was sampled proportionally to its occurrence.

For percentage substratum cover estimates, the following six size categories were used: boulders (>256 mm), cobbles (64–256 mm), coarse gravel (16–64 mm), fine gravel (2–16 mm), sand (0.1–2 mm) and mud (<0.1 mm) (FURSE *et al.*, 1986).

Basic abiotic parameters were also measured during the sampling visits, comprising actual air and water temperatures, pH, conductivity, mean stream width and depth. Monthly precipitation data were provided by the Czech Hydrometeorological Institute.

2.3. Data Analysis

Both abiotic characteristics and biotic metric values (*e.g.*, total abundance, taxonomic richness, relative frequency of main faunal groups) of the sites were compared over the whole sampling period using nonparametric Wilcoxon tests for paired samples (Statistica version 7.0. software; StatSoft, Inc. 2004).

The benthic fauna was divided into two groups: permanent (non-insect fauna: Turbellaria, Crustacea, Nematoda, Enchytraeidae, Oligochaeta, Hirudinea, Crustacea and Mollusca) and temporary (insect larvae: EPT taxa, Megaloptera, Diptera, and Coleoptera, including adults). Species were classified into (i) “r-selection”, “K-selection” and “unspecified-selection” groups (ŠPORKA *et al.*, 2003), and (ii) six feeding groups: grazers/scrapers, shredders, gatherers/collectors, filter feeders, predators and others (MOOG, 1995).

A multivariate approach was applied to highlight biological and functional similarities among samples using CANOCO software (TER BRAAK and ŠMILAUER, 2002). Principal Component Analysis (PCA) was used to analyse spatial (*i.e.* “between sites”) and temporal (*i.e.* “between dates”) variation in community trait selection (*e.g.*, r-strategists *vs.* K-strategists) and feeding habits. The arrays gathering abundance of the mentioned groups in the samples were used for the analyses.

Table 1. Minimum, maximum and mean values of the main abiotic parameters at each site, computed for the whole sampling period. Physical and chemical parameters were compared using the nonparametric Wilcoxon paired test.

	Intermittent site			Perennial site			Wilcoxon test P-values
	MIN	MAX	MEAN	MIN	MAX	MEAN	
water temperature (°C)	4.0	14.7	9.8	4.0	15.1	9.6	0.592
pH	5.7	8.5	7.6	6.2	8.6	7.7	0.208
conductivity ($\mu\text{S} \cdot \text{cm}^{-1}$)	755	930	810	756	1003	861	0.025
dissolved oxygen ($\text{mg} \cdot \text{l}^{-1}$)	9.7	19.5	13.6	9.9	17.0	13.1	0.401
water depth (cm)	2.1	8.6	6.9	4.2	11.8	8.4	–
stream width (m)	1.4	2.6	1.9	1.5	2.9	2.1	–

3. Results

3.1. Abiotic Factors

Basic abiotic parameters from both sites were compared and no statistically significant differences between the main physico-chemical parameters, excepted conductivity were observed (Table 1). Differences between substratum cover at each site throughout the sampling period and also between sites on each sampling date were not statistically significant (Wilcoxon paired test, $P > 0.05$). Consequently, the samples were suitable for comparison, due to proportional sampling design, identical sampling time and effort and minimum differences, throughout the year, in the substratum cover of each stretch sampled. The sites differed mainly in the duration of the dry period.

3.2. Abundance and Taxa Composition of Macroinvertebrate Community

The total number of processed individuals was 77655 and 83 taxa were found. Total abundances and numbers of taxa per sample were compared.

A lower total abundance (Fig. 2) was found at the intermittent site (between 956 and 2653 ind.). Abundance was several times higher (between 1230 and 18500 ind.) at the perennial site throughout the whole sampling period. The highest observed density at the perennial site, in summer 2002, was due to the high number of *Gammarus fossarum*. The following year, after the long-lasting drought, the summer peak did not occur and abundance values were much lower during both the summer and autumn. In the dry year of 2003, total abundances at the perennial site decreased only before the period of drought. In 2002, the year of permanent flow, the summer abundances at the perennial site were higher than at the intermittent site. The total abundances for samples from both the intermittent and perennial sites and for all sampling dates were compared by the nonparametric Wilcoxon pair test and a statistically significant difference ($P = 0.017$) was found.

In contrast, the numbers of taxa (Fig. 2) were higher during the major part of the year at the intermittent site. These varied between 28 (winter 2002) and 37 taxa (spring 2002). At the perennial site, the number of taxa varied from 16 (spring 2002) to 36 taxa (autumn 2003). Numbers of taxa at the intermittent and perennial sites were significantly different (Wilcoxon paired test, $P = 0.027$).

The composition of the macroinvertebrate community (for main faunal groups see Fig. 3) at the intermittent site was heterogenous, without evident trends, except for the strong

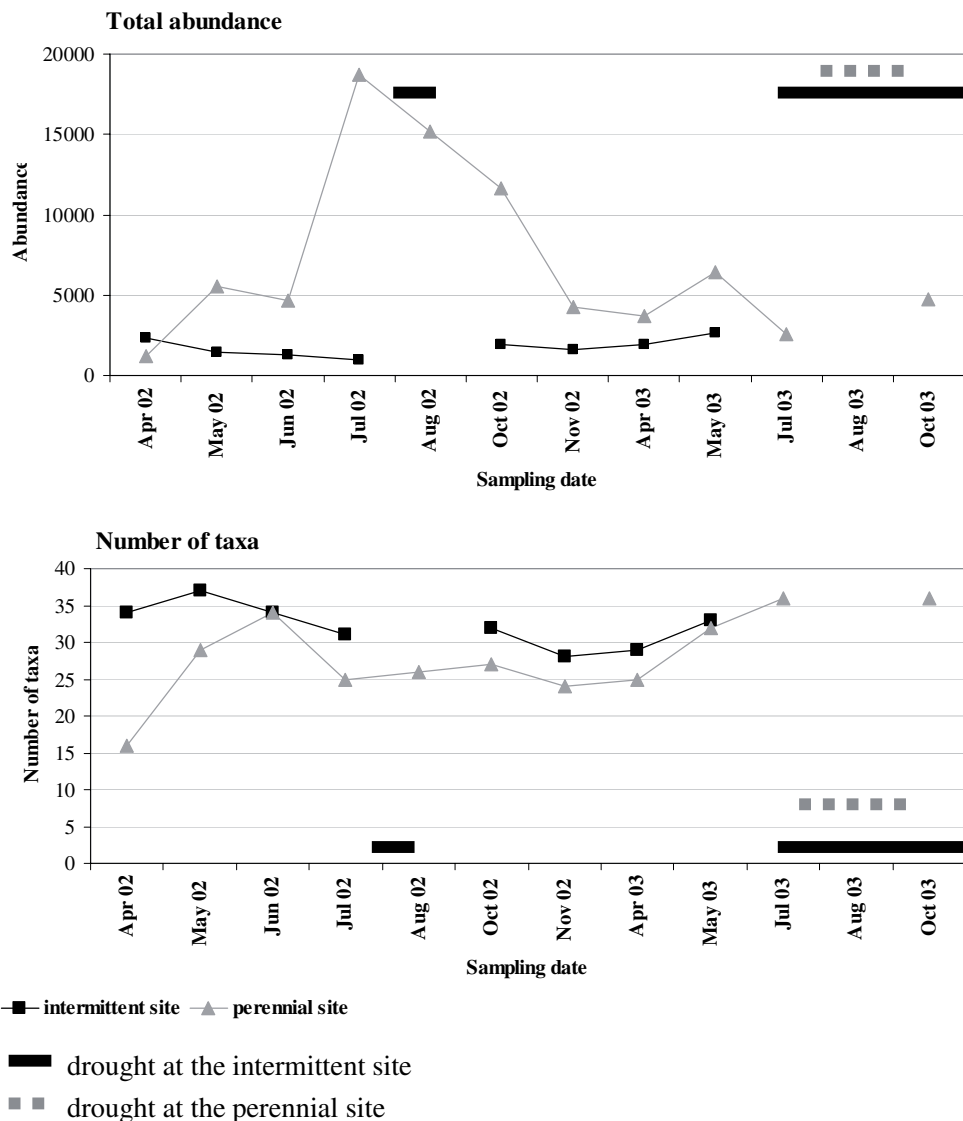


Figure 2. Comparison of the total abundance of individuals and numbers of taxa per sample unit at the intermittent and perennial sites during two sampling years.

increase in dipteran (mainly chironomid) taxa in the pre-drought period. The perennial site was dominated mainly by Crustacea (represented by *Gammarus fossarum*) during the whole first year and the number of dipteran taxa (again represented mainly by chironomids) increased only in the very dry year of 2003, particularly in the spring and summer pre-drought samples. Ephemeroptera, Plecoptera and Trichoptera were represented in all samples; changes in their relative abundance were most probably caused by seasonal effects.

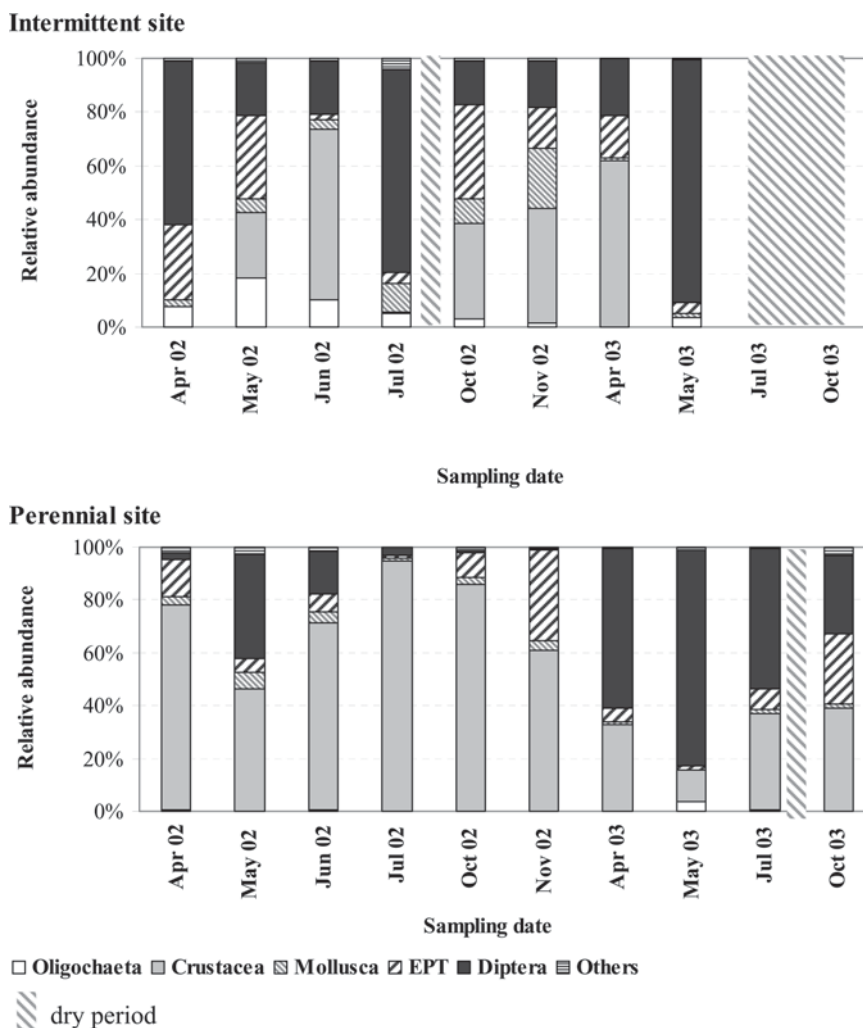


Figure 3. Comparison of the relative abundance values of the main faunal groups (Oligochaeta, Crustacea, Mollusca, Diptera, EPT taxa and others) at the intermittent and perennial site.

3.3. Life Strategies

A comparison of the number of individuals of permanent and temporary fauna formed part of the life strategy analyses. Generally, the permanent fauna (Fig. 4) predominated at the perennial site but their relative abundance was very low in spring 2003, before the 2003 drought, in comparison with spring 2002. The proportion of temporary invertebrate taxa was higher in the intermittent than in the perennial site samples throughout 2002. The proportion of permanent fauna dropped before the dry period at both sites.

Life strategies (number of individuals of r-, K- and unspecified group) for each site were evaluated by PCA (Fig. 5) The 1st axis explained 77.4% and the 2nd axis 22.6% of the known

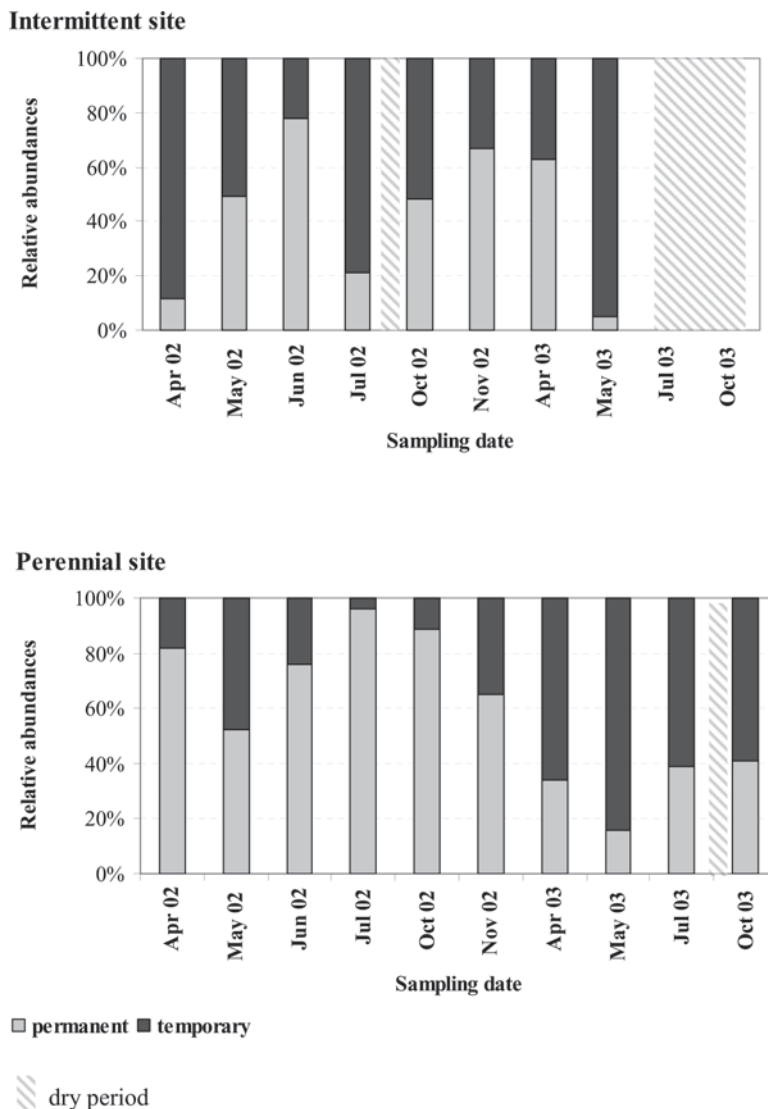


Figure 4. Relative abundance values of temporary and permanent fauna at the intermittent and perennial site.

variability. Most of the samples from the perennial site were situated in the lower part of the ordination graph in the K-strategy factor direction but the samples from the extremely dry year, 2003, were partially shifted in the opposite direction. The samples from the intermittent site were predominantly oriented towards r-strategy factor.

Representation of feeding groups (number of individuals) is evident from Figure 6. Both sites were dominated mainly by shredders and/or gatherers/collectors during most of the sampling period. Nevertheless, before the drought, the relative numbers of gatherers/collectors markedly increased at both sites and in both years. The comparison of feeding groups

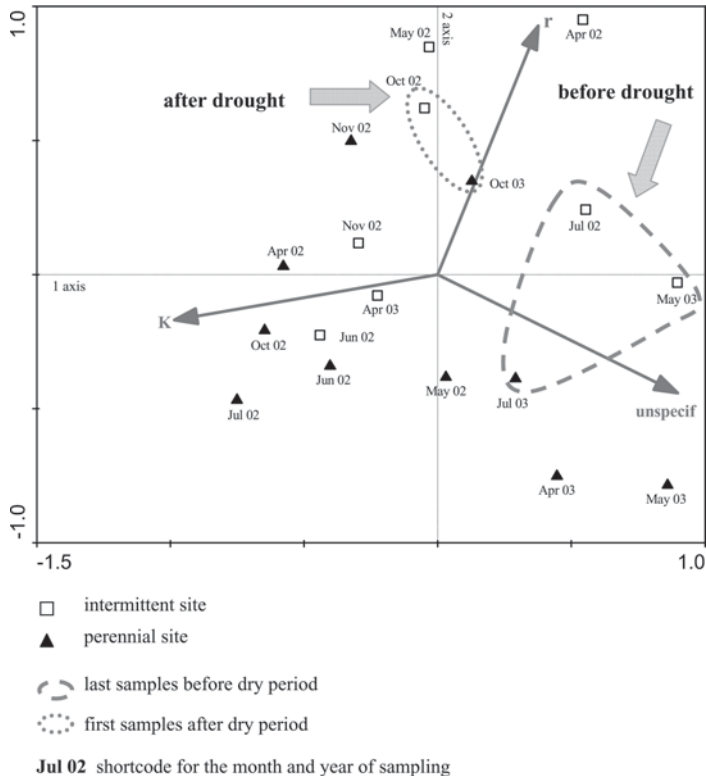


Figure 5. PCA life strategies – comparison of total abundances of K-strategists, r-strategists and an unspecified group at the intermittent and perennial site.

by PCA (the 1st axis and the 2nd axis explained 90.9% of the variation) shows that the sites were very similar and overlapping in this respect (see Fig. 7). Only at the beginning of the dry period, as they are oriented towards gatherer/collector factor, did samples from both sites and years differ from the remaining samples.

4. Discussion

4.1. Abundance and Taxonomic Composition of Macroinvertebrate Community

Throughout the whole sampling period, the lowest abundances of invertebrates were found in samples from the intermittent site (Fig. 2). MILLER and GOLLADAY (1996) and DEL ROSARIO and RESH (2000) also found that total abundances of macroinvertebrates were much higher in perennial streams than in intermittent streams. Total abundances fluctuated considerably at the perennial site in both years, whilst abundances at the intermittent site were relatively stable. The highest abundance observed at the perennial site was in the summer of 2002 and was caused mainly by high numbers of *Gammarus fossarum*. From field observations, this could be connected with the in-stream migration of this species and its concentration, during the dry period, in reaches with residual water.

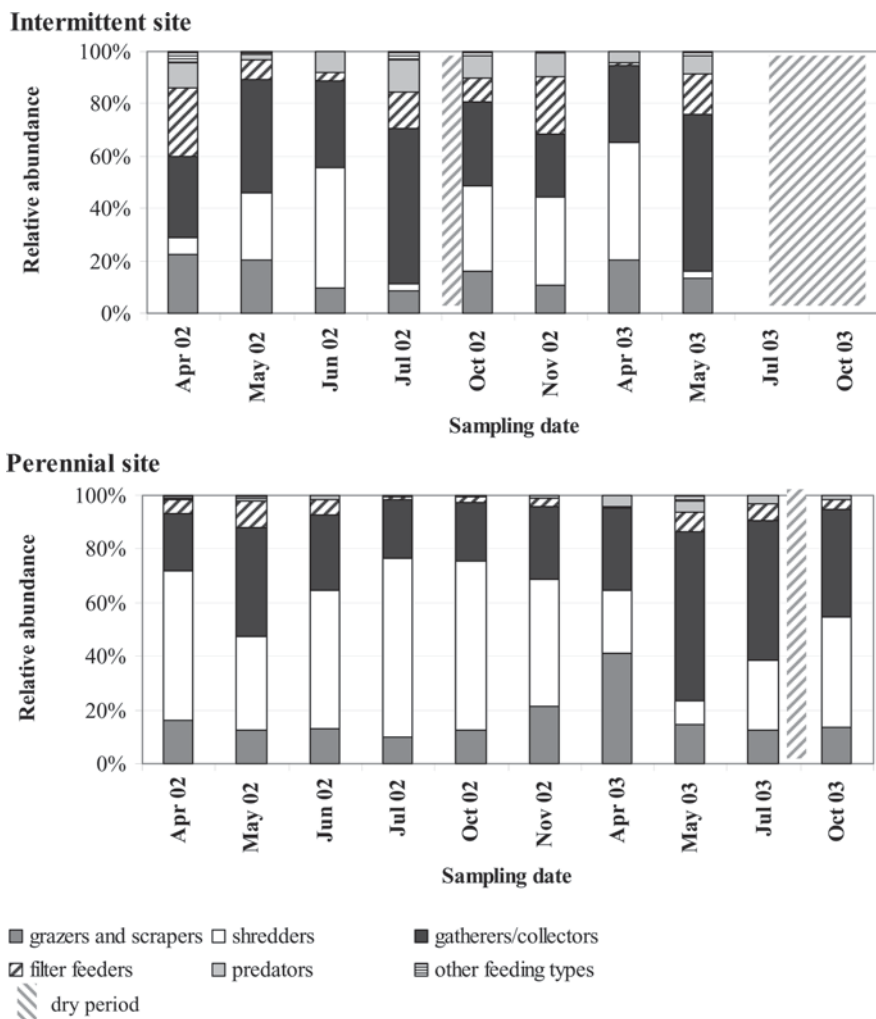
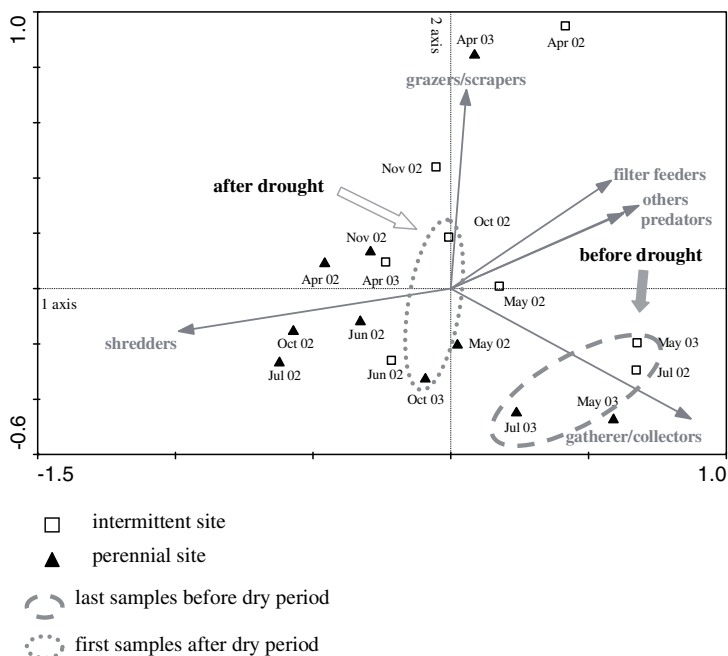


Figure 6. Comparison of relative abundance of six feeding groups (grazers/scrapers, shredders, gatherers/collectors, filter feeders, predators and others) at the intermittent and perennial site.

The number of taxa was relatively stable at both sites (Fig. 2) and rapid decreases did not occur in the samples taken before nor after dry periods. This corresponds with FOWLER (2004), who found that the majority of taxa (95%) recovered already after seven days.

Some faunal groups can be more affected than others, for example Chironomidae and Gammaridae can survive short dry periods (IVERSEN *et al.*, 1978). In the current study, crustaceans (mainly *Gammarus fossarum*) were numerically dominant in most samples, especially at the perennial site (Fig. 3). However, at both sites the number of crustaceans decreased rapidly in samples collected immediately before the drought. As crustaceans were numerous at both sites they were able to compete effectively with other macroinvertebrate groups in recolonization after drought (see the relatively high proportion of this group in



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Figure 7. PCA life strategies – comparison of total abundances of six feeding groups (grazers/scrapers, shredders, gatherers/collectors, filter feeders, predators and others) at the intermittent and perennial site.

after-drought samples, Fig. 3). Chironomids are able to adapt their physiology and life cycles to high water temperature and lower oxygen concentration, as documented by HILSENHOFF (1991), therefore they can rapidly respond to very unfavourable environmental conditions (PIRES *et al.*, 2000).

4.2. Life Strategies

Very low relative abundances of the permanent fauna were recorded at the intermittent site in comparison with the perennial site. An apparent decrease in abundance occurred before the intermittent site dried up in summer (Fig. 4). The permanent fauna dominated at the perennial site in 2002 but the situation was different in 2003 and it was probably influenced by the extremely low precipitation that year.

The permanent fauna can survive drought disturbance in the hyporheal or in the upper or lower flooded stretches (KUBÍČEK, 1988a; DEL ROSARIO and RESH, 2000). Representatives of the temporary fauna can respond quickly to the drought by shortening their life cycles or surviving unfavourable conditions in diapause or as eggs (THORP and COVICH, 1991). Macroinvertebrates, which do not have this predisposition for survival, perish.

Droughts represent a substantial environmental disturbance and the survival of freshwater organisms depends on their life strategies. The intermittent site was periodically disturbed

by drying up and there were relatively more r-selection species there than at the perennial site (Fig. 5). These species are capable of surviving such disturbances and can rapidly colonize new habitats. K-strategists relatively dominated in samples from the perennial site collected during 2002, the year with permanent flow. The more stable and predictable conditions at the perennial site represent better conditions for K-species (SMITH, 1992; TOWNSEND *et al.*, 2003).

Both study sites have similar habitat conditions and probably also food sources. They were dominated by gatherer/collectors and shredders (Fig. 6) but, in the pre-drought period, the relative abundance of shredders always decreased while that of gatherer/collectors increased. There was a reduction of life space with gradual drying up of the streambed and, in relation to such changes in food and habitat resources, feeding groups change too.

4.3. Recolonization and Changes in Community

The rate of recolonization of disturbed habitats, after drying up of the streambed, depends on the length and extent of the drought (KUBÍČEK, 1988b). In the Gránický brook the total abundance values did not decrease markedly after the drought.

Perennial upstream and downstream sections of streams, as well as the hyporheal of the dry stretches, can each serve as refugia and recolonization sources for intermittent reaches (KUBÍČEK, 1988b; DEL ROSARIO and RESH, 2000; FOWLER, 2004). In the Gránický brook, in the first year of study, only a short middle stretch dried up. This dry period lasted only two weeks, so the macroinvertebrate fauna had an opportunity to recolonize this reach by drifting down from upstream sections or by active movement from downstream. However, during the following dry year, rapid recolonization was not possible because the whole brook dried up for six weeks, and the intermittent site for 3 months. In this case, the only recolonization options were from the hyporheal or by compensatory flights of insect adults. Colonization from hyporheal could be the slower of the two mechanisms (KUBÍČEK, 1988b). Generally the rate of colonization is mainly dependent on the substratum porosity and bottom permeability (THORP and COVICH, 1991; RULÍK, 1995; DEL ROSARIO and RESH, 2000).

The Diptera were the numerically dominant group before the drought but they were not dominant in samples after drought. Chironomids were the most abundant group of Diptera species in the Gránický brook, the abundances of other dipterans were negligible. Our findings of Diptera (*i.e.* predominantly chironomids) abundance in after-drought samples (4–6 weeks after re-flooding) corresponds with findings of ELSER (1999) who recorded a very high peak of chironomids on the fifth day after the flow return only, but then their abundance sharply decreased, by more than 50%.

EPT taxa have slower recolonization rates than chironomids (ELSER, 1999), which can be caused by their life cycle. CIBOROWSKI and CLIFFORD (1984) found a slight, linear increase in the number of *Hydropsyche* (Trichoptera) after the return of flow, but even after 12 days this had not stabilised. *Hydropsyche* larvae showed the steep growth of number of individuals in the first 5 days and then the growth was only gradual (CIBOROWSKI and CLIFFORD, 1984; ELSER, 1999). The lower relative abundance of EPT taxa in Gránický brook was found before drying up than after drought at both sites. The higher proportions of EPT taxa after the return of flow, which were found in our samples taken 4–6 weeks after re-flooding, corresponded with the findings of CIBOROWSKI and CLIFFORD (1984) and ELSER (1999) who supposed a gradual growth of abundance after drought. The recolonization rate is specific for each taxon and has to be evaluated in detail.

5. Conclusions

An unstable hydrological regime has significant influence on stream organisms but different natural conditions can generate different community response. Nevertheless, the results of this study are in agreement with observations from other geographic regions. In the case of the Gránický brook, disturbance has no important influence on the number of taxa in both the intermittent and the perennial types of habitat. On the other hand, periodic dry periods caused a significant long-lasting decrease in macroinvertebrate densities at the intermittent site. The composition of macroinvertebrate communities is closely connected to the phenomenon of drying up. Before the drought, the relative abundances of the permanent fauna decreased and relative abundances of r-strategists rose and stayed at a high level in first sample after drought.

Although changes in feeding group composition were found during the pre-drought period (increase of relative abundance of gathering collectors), the exact evaluation of this process needs further and more detailed investigation.

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